

Milestone 1: Foundational Data Analysis Prototype

E-commerce Customer Behavior Analytics System

1. Executive Summary

This document details the foundational phase of the project, focusing on data acquisition and structural auditing. The primary objective was to establish a baseline through descriptive statistics and a comprehensive data quality assessment, ensuring the dataset's viability for subsequent transformation and modeling stages.

2. Dataset Acquisition & Description

The project utilizes a localized dataset detailing customer purchasing patterns, demographics, and membership loyalty tiers.

Source: Local CSV Acquisition ("E-commerce Customer Behavior - Sheet1.csv")

2.1 Data Schema & Variable Definitions

Column Name	Data Type	Definition
Customer ID	Integer	Unique identifier for customer transactions.
Gender / Age	Object / Int	Demographic attributes for segmentation.
City	Object	Geographic location of the customer.
Membership Type	Object	Loyalty tier (Gold, Silver, Bronze).
Total Spend	Float	Cumulative monetary value spent.
Items Purchased	Integer	Total count of items purchased.
Average Rating	Float	Self-reported satisfaction (1-5 scale).
Satisfaction Level	Object	Qualitative sentiment category.

3. Basic Descriptive Statistics

Initial computational analysis provided the following insights into the dataset's distribution:

- **Central Tendency:** Identification of Mean and Median for 'Total Spend' and 'Age' to define the primary customer profile.
- **Variance:** Analysis of the spread in 'Items Purchased' to determine purchasing frequency ranges.
- **Categorical Distribution:** Observation of the frequency across Membership Types and Cities.

4. Data Quality Assessment (DQA)

A critical audit was conducted to identify "dirty" data points requiring remediation in Milestone 2:

- **Completeness:** Detection of null entries specifically within the 'Satisfaction Level' column.
- **Uniqueness:** Identification of redundant rows to prevent revenue double-counting.
- **Validity:** Verification of logical integrity (e.g., confirming no negative spend or unrealistic age values).
- **Anomalies:** Documentation of 0 'Items Purchased' instances that impact future ratio calculations.

5. Preliminary Visual Outputs

Initial visual prototypes were generated to communicate the raw data state, including histograms for spend distribution, pie charts for membership tier share, and count plots for baseline satisfaction volumes.